

WASP-50b

R-1509

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_220126 · created 2026-08-01T15:46:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459605.726 +/- 0.067 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.047
Transit depth (Rp/Rs)^2	1.33 +/- 1.08 [%]
Orbital Inclination (inc)	88.99 +/- 2.91
Airmass coefficient 1 (a1)	1.218 +/- 0.014
Airmass coefficient 2 (a2)	-0.102 +/- 0.0081
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	None
Optimal Aperture	4.84
Optimal Annulus	11.76
Transit Duration (day)	0.043 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.115 $\pm$ 0.047 vs 0.1404 (deviation 0.46 σ)
Duration fit vs published	0.043 d vs 0.0754 d (deviation 0.84 σ)
Mid-time O-C	102.3 min
Depth SNR	2.1
Residual scatter	1.19 %

Light curve

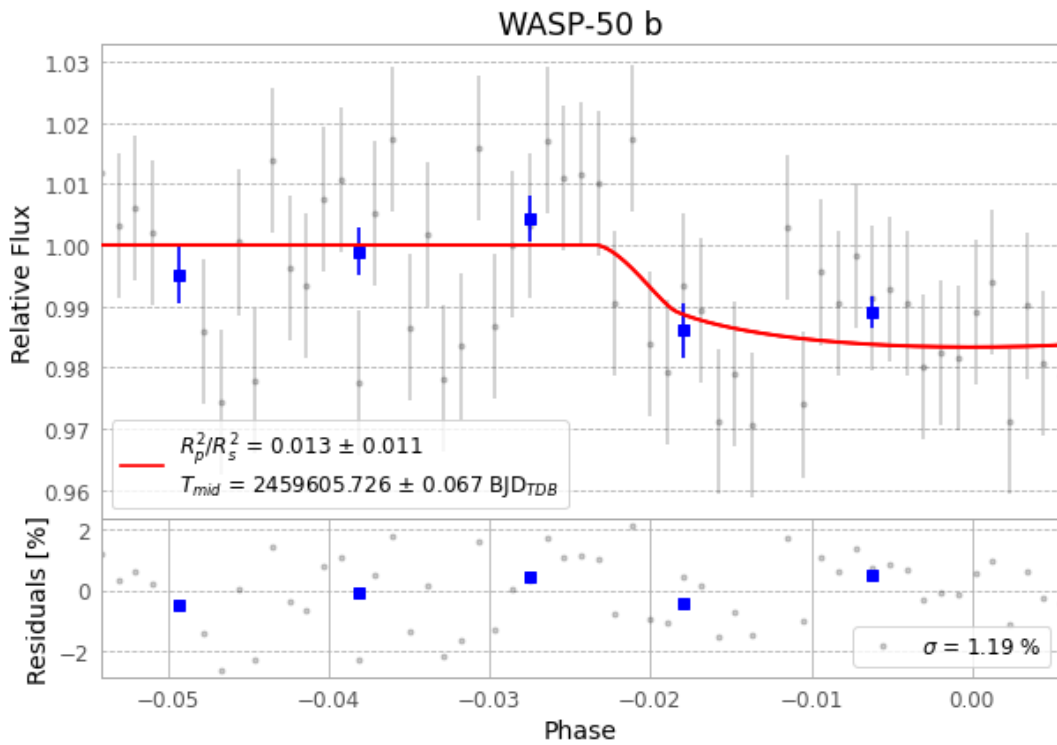


Figure 1 — FinalLightCurve_WASP-50 b_2022-01-25.png (SHA-256 5c3a4467870940f6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [208, 446], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5045b9091c8ada3a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

54 FITS frames (35 MB), WASP-50220126025112.FITS → WASP-50220126053912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-220126.log (197533c7e616...), FinalLightCurve_WASP-50 b_2022-01-25.png (5c3a44678709...), FinalLightCurve_WASP-50 b_2022-01-25.csv (bcb4e68e8bae...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 15:46Z.